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ABSTRACT

Conservation of a complex object, such as an antique clock, demands an interdisciplinary approach. In this case, the object in question is a double-sided, polychrome, wooden, hanging, almost round clock face, measuring approximately 1 m in diameter, with original clock hands and mechanism, dating back to the beginning of the 17th century. The general motivation behind full conservation of this particular object is to bring it back to life so that it can fulfil its main purpose – to indicate time. The paper takes three steps towards this end: the study of the clock and its place in horological history; the conservation of the metal mechanism, gilded clock hands and polychrome, and wooden casing, and the exhibition of a dynamic, working system of the clock face's original, four-century-old components. In this context, the paper examines main restoration issues and discusses the resolutions applied therein.

KEYWORDS: *clock face, horology, 17th century, metal conservation, polychrome wood conservation, exhibition, craftwork, pigment investigation*

Turning back time: Study, conservation, and exhibition of a XVII-century double-sided clock face

INTRODUCTION

It is not common for a piece of art to require research focusing on its artistic and engineering values. In this particular case, investigation and conservation were conducted as a part of the author's master's thesis¹. This allowed for detailed, time-consuming efforts to be put into the project, otherwise limited by scarce resources. This paper discusses the most important results of the work.

The aims of the project were theoretical study required deep research into the general history of clock-making, including a close examination of diffuse and scarce archival sources to re-create the history of the clock. A detailed, laboratory-based study focused on investigating the paint layers. The original primary paint layer and overpainting were identified mainly through a cross-section examination of paint layer samples. Identification of pigments and binders in cross-section samples done by microchemical analysis and comparison of VIS and UV light photography was supported by macro-XRF and SEM-EDS analyses. Altogether, the investigation provided an understanding of the technology and techniques used in the creation of the masterpiece clock, as well as indicate presence of overpainting.

The results of the investigations were the working basis for art conservation decisions. Conservation and restoration of the object involved complex, multistage procedures. The plurality of the used materials and differences in their functions, as well as the state of preservation of the described parts of the object, were the main reason for the applied methodology. Some of the elements needed to be disassembled, whereas others were not because of the risk of permanent damage to the object. The non-demountable parts such as ferrules in sideboards, were treated with the same substances as other metal parts, but differently applied i.e. with gentle application on the metal surface with a syringe, when possible. The extraordinary condition of the clock face presented a unique opportunity to bring it back as a working, moving mechanism

and not just a dull exhibit. Therefore, the work contains remarks on the display and workings of the clock face.

To ensure comprehensive reading of the paper, convoluted and hermetic horological terminology requires some clarification. The word 'clock' means a device for measuring and recording time. This function was probably first fulfilled by simple mechanical bells, without dials or clock-hands, used by monks in the Middle Ages as indicators of prayer time during the day, especially for morning, evening and night time, when sundials were inapt. The word 'klocke' is a Middle Low German and Middle Dutch word meaning 'bell' is a root for the English word. Over time, clocks increased in size and were permanently mounted into church and town halls towers. Decorative dials with numerals and moving clock-hands, visible from afar, appeared on the buildings walls, to avail the entire town society. These however have been powered by the mechanisms at the top of towers, thus can be treated as 'extensions' to the original tower clocks. One clock could have been connected not only with the tower's clock hands, but its drive had a transmission mechanism to the separated clock-faces. These clock-faces were mounted in another parts of the building, such as halls or inside the churches (on walls or under the ceiling), and had their own dials and clock-hands. Because of their cost, sophisticated construction and design, there were a rare sign of wealth and prosperity of its proprietors. Uniqueness and individual characteristics of these objects is the cause for distinguishing between 'clock' and 'clock face' in the paper.

STUDY

HISTORY

The object in question is a double-sided, hanging clock face, consisting of wooden casing and inner clockwork. However, it is not a self-standing clock but a part of a greater mechanism – a tower clock, as mentioned above.

We have no certain evidence of the origins of the clock face. Its most recent history, recorded by



PHOT. 1 A frame from the archival movie *W łogowie zwieria*, showing the clock face hanging in the gallery of the Castle of Lidzbark Warmiński in winter 1945

its current owners, dates back to the end of WWII, when it entered the Warmia and Masuria Museum as a part of the Olsztyn collection. In the course of archival research, the author found a Soviet propaganda movie titled *W łogowie zwieria*² depicting the object in the courtyard of the Castle of Lidzbark Warmiński in winter 1945 (Phot. 1). The movie shows scenes from the former East Prussia cities in the period of the East Prussia Offensive, among them a characteristic courtyard of the castle with the clock face loosely hanging on the second storey of the arcaded gallery.

The inventory markings found on the surface of the clock face tie its provenance to Historischer Verein für Ermland³, a historical society devoted to the study and preservation of Warmia's heritage

and to running a museum in Braniewo.⁴ No more information on the object's history can be found.

A Europe-wide query was sent to many museums and institutions dedicated to heritage technology and the art of northern Europe.⁵ Unfortunately, no institution had an object with similar characteristic features. The clock faces that came close were those found in Oliwa Cathedral in Gdańsk and the Collegiate in Kartuzy, but neither was hanging or had two faces. Therefore, this kind of 'application' for a tower clock seems to be a truly unique one. The greatest help during the reconnoitring of the structure of the clockwork was obtained from Mr. Grzegorz Szychliński, the director of the Museum of Science (former Museum of Turret Clocks) in Gdańsk, to whom the author is very grateful.

Because of a lack of documentation and other historical evidence, an attempt was made to answer two questions: What else can be deduced from the gathered material about the provenance of the clock face? What is its place in horological history?

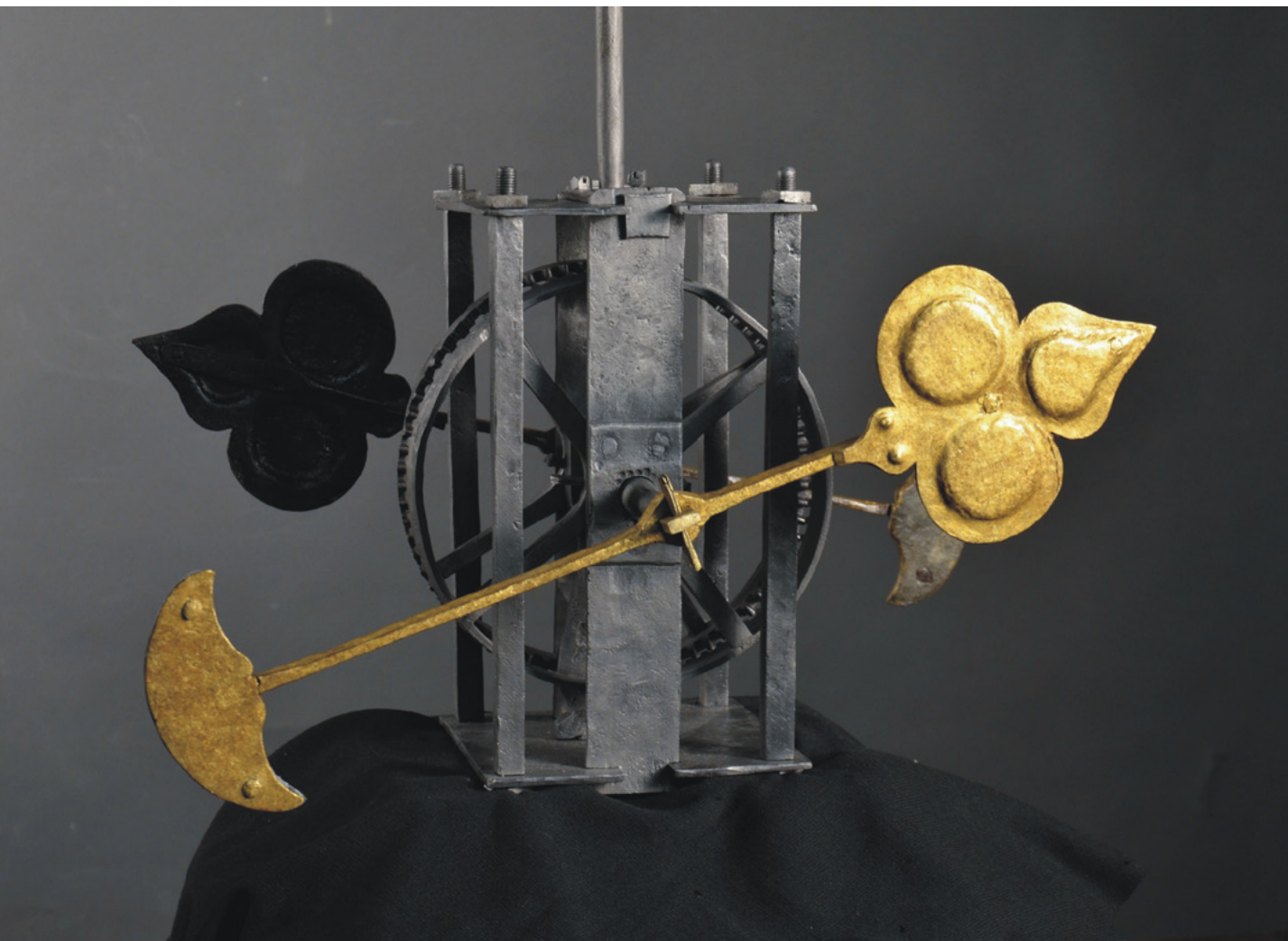
First mechanical clocks have their origin at the end of the 13th century in primitive ringing mechanisms that were used by monks ring bells in cloisters and signal the time for prayer. This mechanical revolution probably started in the cloudy lands i.e. of Great Britain, where sundials, still used in other parts of Europe, were not as functional as in sunny Italy. Those new machines were built of steel and resembled big cages, only with gears and bars inside. Initially, these clocks indicated

only hours. Sometimes, another clock face was designed just for indicating quarter hours. Then, in 1657, thanks to the invention of the first precise time-tracking pendulum by Huygens, clocks began to have another clock hand to indicate minutes. In this period, clocks – which used to be restricted to monasteries and palaces – spread to the houses of nobility and wealthy merchants.

The clock face is equipped with two hour hands – one on each side – strongly suggesting its pre-pendulum period origin. The lack of a minute division on the clock scale is also noteworthy. The inner clockwork is made of hand-forged steel (Phot. 2). Natural imperfections have resulted in the clockwork having only one possible configuration of assemblage,

PHOT. 2 *Clockwork after conservation treatment.*

Picture shot during filming the moving mechanism for documentation by Paula Śwituszek



despite the designed symmetry of the elements. Some of the elements are joined by wedges, whereas others are screwed together. Some of the elements were repaired in the past, including possible rethreading of the bars' screws. However, some 17th-century sources mention screw threads in clock constructions, so it is possible that the original design was made by a technologically sophisticated workshop.⁶ Stylistic analysis of the application of the frugal auricular ornament, strapwork, the motif of laurel leaves, and winged hermai with elongated gothic-like proportions, all date the clock face to the first half of the 17th century. The rounded, slightly vertically elongated shape of the wooden casing suggests a mannerist approach to the form, which is interestingly also present in the dial.⁷

In Northeast Poland, in Warmia, this kind of auricular ornament appeared in woodcarving after 1630.⁸ The framing style, resembling the then prevalent cartouches and designs common in illuminations, and the hermai torsos indicate the object's secular provenance. Hence, analogies of forms should not be drawn from sacral art alone. Ornaments similar to those present on the clock face can be traced back to 17th-century Flemish and Netherlandish books on designs and ornaments then popular in northern Europe, such as *Emblemata Sacra Passionis Salvatoris* (c. 1620) by Lucas Killian, *Neuws Compertament* (c. 1621) by Godtfridt Müller, or *Neues Ziratenbuch* (c. 1640) by Fredrich Unteutsch. Final dating, based on historical and artistic analyses and the technological features of the clockwork, places the origin of the object to ca. 1640.

TECHNOLOGICAL RESEARCH⁹

A close examination of the technological structure allowed for thorough backtracking of the steps of making the object and the spectrum of changes that occurred in the past centuries. The first steps were to create photographic documentation of the object in visible light, registration of UV-induced fluorescence and IR-photography. The two clock faces were carved in linden wood, whereas the sideboard corpus was made of oak wood. When

the clock faces came in for renovation, because of the amount of accumulated surface deposit, it was extremely challenging to judge the state and authenticity of the paint layer. Samples of polychromy were first collected and used in stratigraphic examination; a cross-section of paint layer was carried out, in addition to UV and VIS photographic analysis and XRF spectrometry, to identify the chemical composition. Stratigraphic examination showed that in the majority of the paint layer regions, a second layer of polychromy was present. This was discovered before it became apparent to the naked eye because of the degree of surface soiling. A close examination revealed that different elements were painted in corresponding colours and that the overpainted colours matched the colours beneath. This resulted in the conclusion that the original mending was done with the idea of accurately recreating the colouring. However, this created the new problem of identification of pigments of different compositions but similar cross-section characteristics. SEM EDX and macro-XRF tests were very helpful to solve this problem. The FTIR-ATR method allowed for degradation product detection in blue colour layers.¹⁰ Each layer of paint and gilding was carefully examined. All the pigments used in the original and secondary paint layers were identified. It appeared that parts of the carnation, the silvering (now corroded), the black parts of the clock face, and the angel's hair kept their original colouring, whereas the blue, green, red and gilded parts were re-laid shortly after the object was made. This was probably because of a technological error in the making of the object. The original blue parts were executed with smalt in an oil binder. The degradation of the pigment, resulting in the greying of the paint, was probably the reason for the repainting. The blue part was intended to be blue because it was the sky to the clock's central sun. The overpainting was laid with smalt mixed with lead white, and it has maintained its intensity to this day. The original green and red parts were laid with a water-soluble binder or a very thin layer, possibly resulting in the present scarcity of the original colours on the clock

face. The original gilding was covered by a secondary golden layer. The results of the above investigation were a starting point in the attempt to re-create the intentions behind the first interference in the object's form, evaluate all the previous treatments, and create a working basis for future art conservation decisions.

CONSERVATION¹¹

The greatest challenge of the clock face conservation process was the clock face's complexity and multielement structure. The variety of materials used and their characteristic functions were crucial in taking further steps in conservation. Some parts could be safely dismantled; others, however, were too fragile to be disassembled without risking damage to them. All the objects parts needed to be listed first. To that end, a simple classification scheme of wooden and metal parts was proposed. This scheme was chosen because of the great differences in the methods used in metal and wood conservation and in their storing conditions. In this case, parts with different compositions were treated in separate laboratories.

List of wooden parts:

- 2 linden clock faces (clock face with an ornamental frame)
- 11 oak sideboards
- 1 inner mechanical oak support board
- 2 decorative balls strung on chains

List of steel parts:

- 32 clockwork parts (some with traces of monochromy and gilding)
- 2 gilded clock hands
- 1 polychrome chain
- 2 steel, polychrome ferrules of the oak corpus
- 18 nails and other mounting elements

GENERAL CONDITION OF THE OBJECT

Before the conservation, the object had no support or hanging device to be placed or exhibited on. This

means the entire weight of the object was supported on fragile woodcarvings, with different pieces barely stacked together. The greatest damage was probably done during WWII because of mishandling and exposure to harsh environmental conditions.

The wooden casing – the clock faces and the sideboards – had been attacked by insects in the past, which was apparent from the tunnels, wood holes, and organic deposits. The ornament, because of poor handling conditions, was broken in numerous spots. The loss of the original paint layer and gilding was also apparent. The outer surface was covered in grime and dust and had been subjected to abrasion, scratches, and stains. The silver elements were corroded. The paint layer was cracked and fragile, especially the green paint layers, probably because of past overdrying of the wood. The ornament had been nailed to the object during an early restoration, which resulted in the boards cracking endwise and athwart. Additionally, bar clamp markings were present on the wood and paint layer. The metal parts were mostly corroded, and the clock hand's fragile gilding was damaged because of steel corrosion.

MATERIALS AND METHODS

After photographic documentation of the object, the next step was dismantling the clock face into parts and keeping the wooden parts in a controlled environment. They were kept at 60% RH, whereas the metal parts were kept at 30%–40% RH in room temperature.

WOODEN PARTS

After dismantling, each clock face was first cleaned of the primary dirt deposit. The remains of insect habitations were removed, and entomology samples were collected for research. Then, the detailed surface cleaning stage could begin. Different types of rubber of various hardness were used for surface dirt removal, including an electric eraser and brushes of natural and nylon hair. Some places, for example, in parts of the oil-bound paint carnation, were cleaned with a cotton bud moistened



PHOT. 3 *Clockwork after conservation treatment.*

Picture shot during filming the moving mechanism for documentation by Paula Śwituszak

with Marseille soap foam because of its neutral pH. Stains on versos were removed with the use of water-soaked cotton buds. A similar method was used for the blue parts to remove stains by gently rolling over them with a small, water-soaked cotton bud. The gilding was cleaned with synthetic ox gall, which was rubbed on with a water-soaked cotton bud. Because of the great amount of deposit, some spots required two or three recleanings or the use of a different type of rubber. Some paint layers and gilding blind flaking had been glued on by a 7% fish glue solution. The cracks in the clock face were glued together by insertion of 1–2 mm thin piece of linden wood, covered by a layer of glue based on polyvinyl alcohol (PVA), which has good wood-binding properties and also creates strong, yet flexible layers. The glued spots were reinforced with two blocks of wood, 2,5 cm x 3,5 cm x 0,3 cm, glued in the upper part of the clock face's backside with PVA.

Reconstruction of the missing ornamental framing parts was completed. The work began with the preparation of a smooth contact surface and gluing of three wooden blocks with Vinawil 59 glue reinforced with two 3 mm beech pins.

Missing ornament parts were carved out of linden wood, following the original design of the single surviving corresponding part of one of the clock faces. Using the same method, other parts of the ornament were reconstructed. Before the next step, the entire surface of polychrome parts was covered by LeFranc & Bourgeois Retouching Varnish Spray Extra Fine, to protect the original parts. Then, the remaining cavities were filled in with epoxy raisin – Araldite HV 427 mixed with hardener HV 427 and a bit of earth pigment. Fillings were smoothed with sandpaper and scalpel. Missing parts of the grounding were filled in with Viennese filler,¹² smoothed with sandpaper and a water-soaked cotton bud. Reintegration of the paint layer was completed using Rembrandt watercolours. The reintegration of the paint layers was completed for the green and red layers because they were the most damaged areas. Second, the fillings were coloured. Then, the surface of watercolour pinpointing was covered with a protective layer of spray acrylic satin LeFranc & Bourgeois varnish, not only because of its stability against light exposure, but also because of its not glistening visual properties, which best fit the appearance

of clock-face. Aesthetic paint layer and gilding reintegration were achieved with pigments mixed with 20% acrylic binder Paraloid B27 in PGME, with PGME + DPGME thinner in a 4-to-1 proportion.¹³ In regions where a primarily metallic foil was attached, silver and gold powder were used. The final protection layer was completed using the same satin spray varnish as before, and parts of the carnation and hair were gloss varnished by brush (*Royal Talens Amsterdam, Glossy Acrylic Varnish, no. 114*). The clock face's back, support mechanism, and wax paste was thinly applied to sideboard's back (Antiquaire Starwax, vaisz, közép tölgy), creating a hydrophobic barrier (Phot. 3).

The sideboard corpus was disassembled and cleaned in a similar way to that used for both sides of the clock face. Dust, stains, and surface deposit were removed in the same way, mainly with the use of rubber and a water-soaked cotton bud. The sideboards were originally glued together with glutin glue, whose remains were removed with a brush, scalpel, and hot water-soaked cotton bud. The next step was filling in wood cavities that had resulted from insect habitation. This was done with epoxy resin – Araldite HV 427 and HV 427 hardener. Cracks, detachments, and broken-off pieces were treated with Vinavil NPC Stella Bianca glue, and bar clamps, and rubber bands and, in small damaged areas, art clips were used.

METAL PARTS¹⁴

Nails, that detachment was possible, were treated differently from metal elements, whose removal would have damaged parts of the clock face, such as ferrules and side angles. The impossibility of removing the metal parts without compromising the character of the original structure made the conservation process complex. Ferrules that removal would risk damage to the casing were cleaned with a scratch brush with a fiberglass tip, and a 4% disodium edetate solution was used where monochromy appeared. Finally, ferrules were covered in 10% tannin ethanol solution and 10% Paraloid B-44 xylene solution.

Corrosion was removed from the nails mechanically. This was supported with a 10% orthophosphoric acid solution. In the case of the clock hands, corrosion products and deposits in the ungilded parts were removed mechanically with abrasive bonded fabric and a scratch brush with fiberglass tip. The gilded surfaces were chemically cleaned with 4% disodium edetate solution. Then, the nails and the clock hand's entire surface were degreased with acetone and covered with 10% tannin ethanol solution. A surface anticorrosion protective layer was introduced using 10% xylene Paraloid B-44 solution with 1% additive of the photostabilisers Tinuvin 292 and Tinuvin 171.

The lead counterweight was cleaned with acetone. The paint layer abrasions were reintegrated chemically with Patynol CYN F. Then, the missing counterweight was reconstructed from a lead plate. The piece was cut to the desired shape and attached to the clock hand with copper rivets. The clock hands gilding was reintegrated with gold powder with 20% Paraloid B-72 binder.

The mechanism was dismantled into thirty-two basic parts. Mechanical cleaning trials (involving abrasive bonded fabric, steel brush, scratch brush with fiberglass tip), conducted with some elements such as mechanism bars or axis support, had promising results. However, subsequent trials indicated that the methods used were insufficient for surfaces that were more corroded. Hence, the sand blasting method was used to clean all the parts. The parts from which the corrosion deposit were removed were degreased with acetone, covered with a 10% tannin ethanol solution, and seasoned in 60°C for a week to stabilise the iron corrosion process. Next, the metal surface was worked on with abrasive bonded fabrics of various levels of hardness. Finally, all parts were covered with 10% xylene Paraloid B-44 (DOW) solution with 1% additive of the photostabilisers Tinuvin 292 and Tinuvin 171. Reassembly of the mechanism required the use of grease in the axis support area. Additionally, a brass distance ring was used to stabilise the movement of the clock hands.

The chain, similar to the gilded parts and monochrome shielding, were chemically cleaned using a 4% disodium edetate solution. The surface was degreased with acetone and later soaked in 10% tannin ethanol solution. A surface anticorrosion protective layer was introduced. The wooden decorative balls that had been gilded and silvered using a mixture technique were cleaned gently with synthetic ox gall. Retouches were made with gold and silver powder with 20% Paraloid B72 in PGME and solvent PGME + DPGME, as mentioned earlier.

REASSEMBLY

Because of the poor condition of the object before treatment, only in the end – that is, after regluing and stabilisation of all the parts of the clock face – did we realise that the clock face was not of the highest possible precision, even though the sideboards and clock face had been made neatly. Both the clock face and sideboards, when put together, fit very well. However, for the final reassembly, all the parts had to be forcefully put together: the sideboards were hammered into one of the clock faces, and then the mechanism was put back inside the corpus with its support so that the sides of the entire corpus were a bit deformed. This resulted in some of the sideboards not fitting the second clock face. Two angle brackets were finally added on the sides to hold the structure together. The angles were mounted a bit higher – in a place determined by the deformed corpus. Because of the irregularities in the structure, the bottom part of it was missing two of the original sideboards. These had probably fallen off because of widening of the distance between the two clock faces at the bottom level. To stabilise the object and prevent future deformations, we fit the sideboards to their corresponding spots, and one of the sideboards and mechanism support board were shortened by around 0,5 cm. The two missing sideboards were remade. All the wooden elements were mounted using the original nails in their original locations. During the final reassembly of the clock face, the corpus sideboards were not glued together, as had originally been done. Such a procedure



PHOT. 4 Clock face after conservation treatment.
Paula Śwituszak

would have made the reassembly even more complex and made the exact fitting of the parts impossible. The original angles prevented the distance between the two clock faces from being changed at the top and bottom levels. To increase the stability of the overall structure and prevent the sideboards from falling out, an ochroma pad was installed. To stabilise the hanging clock face, the original bar element connecting the side of the clock face to the wall was reinstalled. A wooden support for the clock suspension was built (Phot. 4).

EXHIBITION

As our Europe-wide inquiries showed, a complete clock face with its original mechanism is a unique museum specimen. This qualifies the specimen for careful exhibition and analysis. After consultations,



PHOT. 5 A small engine and control platform are visible on one side of the frame from which the clock face is suspended.
Paula Śwituszak

in agreement with the curator of the Warmia and Masuria Museum in Olsztyn, Mr. A. Rzepołuch, it was decided that after the conservation, the object would be exhibited in the Castle of Lidzbark Warmiński, where it was found in 1945.

As is well understood, an object composed of wooden and metal parts must be exhibited in proper and stable environmental conditions. It is also important to adequately establish the suspension height. For the best functionality and visibility of the clock face, the suspension height should have a proportion of 10:1 to 12:1 to the diameter of the clock face.¹⁵ In this case, for a clock face dial length of 56 cm, the suspension height should be 5.6 m to 6.72 m. However, the suspension height can be adjusted for the exhibition capabilities of the museum. These numerical values are suggested only to ensure clear visibility of the object.

The clock's drive proposed by author was based on the idea of a combined system of a control box and a stepper motor designed by an electromechanical company to ensure the clock was always working during the exhibition. To test the possibility of using the original mechanism, a toy drive was made based on a control platform and servomechanism connected to the axis of the mechanism by a steel joint (Phot. 5). The mechanism can simply be plugged in and programmed to turn the clock hands every 30 minutes to indicate every full hour and half hour. This can be easily regulated by using an Arduino environment, to ensure that the clock works like it did in the beginning of the 17th century.

The exhibition will also include historical discussions conducted by a multimedia panel, a movie on the workings of the inner parts of the clock, archival pictures from Lidzbark and photographic documentation of the state of the object during the stages of the overall conservation process. A 3D model of the clock face may also be made to allow the visually impaired to 'see' the object through touch. In the model, every colour will have a slightly different surface texture.

CONCLUSIONS

This endeavour has presented diverse opportunities to use the interdisciplinary approach to conduct holistic treatment and presentation of a unique object.

The work resulted in conservation and close study of the object from historic, artistic, and technological perspectives. Because the clock had to be repaired and the clock face repainted, the object must have been in use for a long time. The current colours are not original, but all the past repairs were made with the use of the best materials of the highest quality. The colours were applied correspondingly to the previous colouring, which testifies to the great care taken by the previous owners. The indications of these repairs were documented along with the entire process of study and restoration of the clock face.

The most important accomplishment of the undertaking was the creation of a unique museum specimen with outstanding exhibition values. The specimen has already been acknowledged by 19th-century historians and museum curators to retain its full functionality as a colourful 17th-century double-sided clock face. Today, in 21st century, we are not only able to recognize the timeless values encapsulated in the fabric of a museum specimen, but also bring it back to life in its full functionality, recreating the original experience of the first users. Our project illustrates that this opportunity, brought up by the popularisation of autonomous programmable electronic systems, although not commonly recognized in the field of art conservation, while grasped and endorsed opens new dimensions for the exploration of heritage.

[1] Two distinct sections of the MA thesis were overseen separately. Conservation and artistic part, provided by Joanna M. Arszyńska PhD and theoretical-research part provided by prof. dr hab. Justyna Olszewska-Świetlik: 'From Groundwork to Exposition – The Unique Clock Face from Collection of Muzeum Warmii i Mazur w Olsztynie', Nicolaus Copernicus University in Toruń, 2018.

[2] W łogowie zwieria, [online video], [1945], <https://www.youtube.com/watch?v=vUSdcypsoNc>, (accessed 17 February 2019).

[3] The letters 'HVE 137' were interpreted as the abbreviation of 'Historischer Verein für Ermland', an association functioning from 1856 to 1945. It was restarted in 1955/1956 in Westphalia. According to information from its official website (<http://his.ermlandfamilie.de/>), the entire previous archive was lost during World War II.

[4] S. Archemczyk and A. Szolc, *Braniewo*, Olsztyn, 1995, pp. 223-224.

[5] Some of the institutions queried were Muzeum Nauki Gdańskiej (previously Muzeum Zegarów Wieżowych w Gdańsku), National Museum in Warsaw [Muzeum Narodowe w Warszawie], National Museum in Poznań [Muzeum Narodowe w Poznaniu], Muzeum im. Przypkowskich w Jędrzejowie, Deutsches Uhrenmuseum, Wien Musuem, Uhrenmuseum Beyer Zürich, Uhrenstube Aschau, Museum für Turmuhren und Bratenwender, British Horological Institute, Comtoise Uhren Museum, Musée de l'horlogerie Department of Mechanical Engineering, National Technical Museum in Prague, and Turmuhrenmuseum Mindelheim.

[6] G. Capriglia, *Misura del tempo cioè tratta-to di horologij da ruota di tre ordini*, 1665.

[7] Clock face dimensions: height – c. 90 cm, width – c. 82,5 cm, dial: c. 56 x 54 cm.

[8] J. Samek, *Polskie rzemiosło artystyczne. Czasy nowożytne*, Warszawa, Wydawnictwa Artystyczne i Filmowe, 1984, p. 138.

[9] Detailed description in MA thesis: P. Śwituszek, 'From Groundwork to Exposition – The Unique Clock Face from Collection of Muzeum Warmii i Mazur w Olsztynie', Nicolaus Copernicus University in Toruń, 2018.

[10] P. Śwituszek, et al., 'Clock Face from the 1st half of 17th century from Warmia and Masuria Museum in Olsztyn collection – historic, stylistic, technological and conservation issues', [Tarcza zegarowa z 1 poł. XVII w. z Muzeum Warmii i Mazur w Olsztynie – zagadnienia historyczno - stylistyczne oraz technologiczno – konserwatorskie], *Acta Universitatis Nicolai Copernici*, vol. 49, 2018, pp. 231-260.

[11] Detailed description in MA thesis: P. Śwituszek, 'From Groundwork to Exposition – The Unique Clock Face from Collection of Muzeum Warmii i Mazur w Olsztynie', Nicolaus Copernicus University in Toruń, 2018.

[12] Filler based on polyvinyl alcohol, methylcellulose, venetian turpentine, mastic in turpentine oil, and pigments.

[13] PGME: 1-metoxo-2-propanol, 98,5%, extra pure; DPGME: Dipropylene glycol monomethyl ether, mix of isomers 97%

[14] Metal conservation support was provided by Alina Tomaszewska-Szewczyk, PhD, from the Department of the Conservation-Renovation of Architecture and Sculpture, Nicolaus Copernicus University in Toruń

[15] W. M. A. Podwapiński, *Zegarmistrzostwo*, T. IV, Niepokalanów 1950, p. 252.

[1] Archemczyk, S. and Szolc A., *Braniewo*, Olsztyn, 1995.

[2] Podwapiński, W. M. A., *Zegarmistrzostwo*, T. IV, Niepokalanów, Wydawnictwo oo. Franciszkanów, 1950.

[3] Samek, J., *Polskie rzemiosło artystyczne. Czasy nowożytne*, Warszawa, Wydawnictwa Artystyczne i Filmowe, 1984.

[4] Śwituszek P., et al., 'Clock Face from the 1st half of 17th century from Warmia and Masuria Museum in Olsztyn collection – historic, stylistic, technological and conservation issues', [Tarcza zegarowa z 1 poł. XVII w. z Muzeum Warmii i Mazur w Olsztynie – zagadnienia historyczno - stylistyczne oraz technologiczno – konserwatorskie], *Acta Universitatis Nicolai Copernici*, vol. 49, 2018.

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